

variance $\propto$ how much the model's prediction would change if trained on a completely different training dataset

inference $\propto$ understanding relationships between features

bias $\propto$ unaccounted factors

prediction $\propto$ focus only on the outcomes

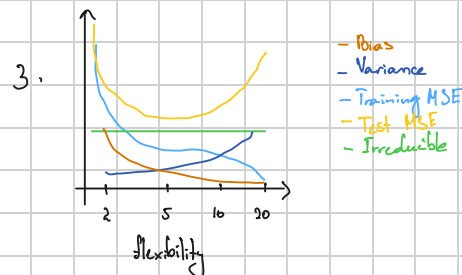
1. flexible vs inflexible performance

- a) flexible $\rightarrow$ opposite to the next one
- b) easy to overfit $\rightarrow$ inflexible
- c) flexible $\rightarrow$ won't fit into standard models
- d) inflexible $\rightarrow$ lower variance

2. a) $n=500$ $p=3$
regression
inference

b) classification
 $n=20$ $p=13$
prediction

c) regression
 $n=52$ $p=3$
prediction



Bias gets lower as flexibility increases, but, instead, variance gets higher due to possible overfitting. The training MSE keeps getting smaller due to the same reason while when the set is switched, the MSE gets higher than the training one. The training MSE can outperform the irreducible error that way, but the testing MSE cannot use it down.

Irreducible Var(E): lowest achievable test MSE among all possible methods

$$\text{Test MSE} = \text{bias}^2 + \text{variance} + \text{irreducible error}$$

4. a) 1) Fraud detection
response: 1 or 0
predictors: device, amount, time, location
prediction

2) Disease detection
response: 1 or 0
predictors: temperature, bloodwork, ECG etc
prediction
if clinical if research

3) coin in elections prediction
response: 1 or 0
predictors: popularity, difference from public opinion, favorability

b) 1) Stock market
percentage change
response: percentage change
predictors: previous performance in the similar days or previous similar days
prediction

2) House Cost Prediction
response: amount
predictors: area, crime, school ratings, roads, parks
prediction

3) Currency Value
response: amount of increase or decrease
predictors: Central Bank interest, recent amount of selling or buying gold, debt, reengagement
prediction

c) Recommendation systems based on groups; Disease detection; store Selling Performance in different areas

5. Flexible models perform better on large datasets (better high variance) where prediction is preferred. Overfitting is the main issue. Best for complex tasks

Less flexible are better for small datasets (bias prevents overfitting) where inference is more important

6. Parametric estimates coefficients of same function, better for sets that follow specific curves but bad for wrongly selected curve

Non parametric doesn't assume any shapes and works better on non-linear sets. However it's more prone to overfitting

7. Euclidean Distance

$$d = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} = (x_i - x_j)^2 + (y_i - y_j)^2$$

	1	2	3	4	5	6
1		3.6	3.6	2.8	1.4	2.4
2			3.7	3	3.1	1.7
3				1.2	2.2	
4					1.7	1.4
5						2.2
6						

a)

	0	1	2
1		3	
2		2	
3		110	3.16
4		$\sqrt{5}$	2.23
5		$\sqrt{2}$	1.41
6		$\sqrt{3}$	1.73

b) Green, the closest
c) 1 green, 2 red, red

d) Smaller is better $\rightarrow$ more flexible boundary.
High just averages everything out $\rightarrow$ linear boundary.